

ASTORIA, OR, USA

WMO: 727910

Lat: 46.157N

Lon: 123.883W

Elev: 9

StdP: 14.69

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 94224

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	28.6	31.1	16.7	12.8	34.2	22.9	17.2	35.9	32.0	52.8	27.9	52.3	5.6	80	0.578

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.7	77.1	63.5	72.4	62.1	69.3	61.1	65.2	74.3	63.5	70.3	62.1	67.9	10.9	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.5	81.8	67.3	60.5	78.6	66.0	59.4	75.8	64.7	30.0	73.9	28.7	70.3	27.7	67.7	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
24.0	19.9	17.7	DB	23.8	88.4	3.6	4.1	21.1	91.3	19.0	93.7	17.0	96.0	14.3	99.0
			WB	22.0	70.0	3.9	1.7	19.2	71.2	16.9	72.3	14.7	73.2	11.9	74.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.7	44.2	44.3	46.2	48.9	53.4	57.4	60.8	61.5	59.2	53.1	47.6	43.5
	DBStd	7.81	5.16	4.85	4.24	4.14	4.15	3.46	2.90	2.87	3.91	4.53	5.47	5.46
	HDD50	896	188	165	131	66	11	0	0	0	0	20	107	208
	HDD65	4870	645	580	581	484	361	232	135	115	181	368	522	666
	CDD50	1530	8	5	15	33	118	221	336	357	276	118	35	8
	CDD65	23	0	0	0	0	2	3	5	7	6	0	0	0
	CDH74	299	0	0	0	9	32	44	58	79	72	5	0	0
	CDH80	75	0	0	0	1	9	12	17	22	14	0	0	0
	WSAvg	7.5	7.9	7.7	8.0	7.7	7.4	7.6	7.9	7.0	6.5	6.7	7.7	8.4
	PrecAvg	68.4	10.4	7.4	7.4	5.2	3.3	2.4	1.0	1.4	2.7	6.1	10.7	10.7
(15) Precipitation	PrecMax	92.7	24.1	21.9	15.3	10.1	7.2	5.5	4.4	5.4	7.3	13.5	19.6	20.4
	PrecMin	41.6	0.7	1.3	1.0	1.7	0.4	0.6	0.0	0.0	0.0	0.5	1.5	2.7
	PrecStd	12.2	4.7	3.9	2.9	2.0	1.4	1.3	1.0	1.2	1.9	3.2	3.9	3.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.6	60.3	65.8	72.9	78.0	79.8	81.5	82.6	81.7	73.2	63.6	58.9
		MCWB	52.8	49.6	52.9	57.8	61.6	65.2	67.1	67.0	63.9	59.3	56.9	55.3
	2%	DB	55.4	56.5	59.1	63.5	68.7	70.7	72.9	74.2	74.9	66.8	59.9	55.5
		MCWB	51.3	49.3	50.9	54.1	58.5	61.5	63.6	64.1	62.7	57.1	55.5	52.8
	5%	DB	53.5	53.6	55.6	58.7	63.3	66.4	69.4	70.3	69.9	63.1	57.8	53.3
		MCWB	50.7	49.3	50.1	51.8	56.7	58.9	61.8	62.5	60.9	57.1	54.2	50.8
	10%	DB	51.8	51.6	53.2	55.5	60.3	63.8	67.1	68.0	66.6	60.7	56.1	51.5
		MCWB	49.2	48.3	49.0	50.3	54.6	57.5	60.5	61.2	59.7	56.1	52.9	49.2
	0.4%	WB	55.2	54.1	54.9	58.8	63.3	65.9	67.7	67.6	66.5	62.4	58.9	55.7
		MCDB	56.9	56.3	60.3	70.6	74.8	78.6	78.9	80.6	76.8	67.3	61.5	57.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	53.0	52.0	52.9	55.0	59.4	62.1	64.3	64.8	63.8	59.9	56.7	53.3
		MCDB	54.2	54.0	56.7	60.9	66.9	70.1	71.9	73.2	70.9	63.9	58.9	54.9
	5%	WB	51.3	50.4	51.2	52.9	57.0	59.9	62.5	63.3	62.2	58.1	55.2	51.4
		MCDB	52.8	52.5	54.5	57.1	62.7	65.2	68.2	69.5	68.0	61.6	57.2	52.7
	10%	WB	49.8	48.4	49.9	51.2	55.1	58.3	61.2	62.0	60.7	56.7	53.5	49.6
		MCDB	51.5	50.9	52.9	54.6	59.3	62.8	66.1	67.1	65.6	59.9	55.3	51.4
(35) Mean Daily Temperature Range	5% DB	MDBR	10.5	12.2	12.9	13.3	12.9	12.5	12.6	13.7	15.8	14.4	12.1	10.1
		MCDBR	11.6	15.6	17.9	20.0	21.1	18.9	18.2	19.9	23.7	19.2	13.2	10.5
		MCWBR	8.7	10.3	11.3	12.1	12.8	11.1	10.6	11.6	13.3	11.7	9.2	9.0
	5% WB	MCDBR	9.6	11.5	14.5	16.6	18.1	17.1	16.5	17.9	19.1	14.5	9.7	9.5
		MCWBR	8.6	8.8	10.0	10.7	11.3	10.4	9.9	10.7	11.3	9.9	7.9	9.1
(40) Clear-Sky Solar Irradiance	taub		0.330	0.319	0.336	0.367	0.387	0.367	0.352	0.357	0.346	0.336	0.327	0.322
	taud		2.485	2.554	2.519	2.394	2.358	2.440	2.510	2.493	2.530	2.550	2.523	2.495
	Ebn at Noon		244	275	285	283	279	285	288	283	277	264	244	234
	Edh at Noon		22	24	29	35	38	35	32	32	28	24	21	19
(44) All-Sky Solar Radiation	RadAvg		357	628	939	1361	1681	1837	1966	1691	1313	776	421	298
	RadStd		53	80	110	119	149	155	155	133	110	87	53	41

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+45

Nomenclature: See separate page